

Day 5: SQL Notes Basic to Advanced

Today's Topic: SQL Basic Queries : Select Statement

- SELECT Statement
- SELECT UNIQUE
- SELECT DISTINCT
- SELECT COUNT
- SELECT IN
- SELECT DATE
- SELECT SUM
- SELECT NULL

5.1 SELECT Statement

The SELECT statement is one of the most fundamental commands in SQL (Structured Query Language). It is used to retrieve data from one or more tables in a relational database.

- Retrieves data from one or more database tables.
- Allows filtering, sorting, and grouping of results.
- Can specify particular columns to display.

Syntax:

```
SELECT column1, column2, ...  
FROM table_name;
```

SELECT: Specifies the columns you want to retrieve.

FROM: Specifies the table from which to retrieve the data.

Example:

1. Select All Columns:

```
SELECT * FROM employees;
```

This retrieves all columns from the employees table.

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2. Select Specific Columns:

```
SELECT first_name, last_name  
FROM employees;
```

This retrieves only the first_name and last_name columns from the employees table.

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3. Select with Conditions:

```
SELECT first_name, last_name
FROM employees
WHERE department = 'Sales';
```

This retrieves the **first_name** and **last_name** columns for employees who work in the 'Sales' department.

4. Select with Sorting

```
SELECT first_name, last_name
FROM employees
ORDER BY last_name ASC;
```

This retrieves the **first_name** and **last_name** columns and sorts the results by **last_name** in ascending order. Use **DESC** for descending order.

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5. Select with Aggregation

```
SELECT COUNT(*) AS total_employees
FROM employees;
```

Select with Aggregation

5.1 SELECT UNIQUE

The UNIQUE keyword is used to ensure that the values retrieved are distinct. However, it is not standard SQL and is mostly used in Oracle databases. The standard SQL keyword for this purpose is DISTINCT

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Example (Oracle):

```
SELECT UNIQUE column1
FROM table_name;
```

5.2 SELECT DISTINCT

The DISTINCT keyword is used to remove duplicate rows from the result set, ensuring that each row is unique.

Syntax:

```
SELECT DISTINCT column1, column2, ...
FROM table_name;
```

Example:

```
SELECT DISTINCT department
FROM employees;
```

This query retrieves a list of unique departments from the employees table, removing any duplicate entries.

5.3 SELECT COUNT

The COUNT function returns the number of rows that match a specified condition. It is often used with the GROUP BY clause.

Syntax:

```
SELECT COUNT(column_name)
FROM table_name
WHERE condition;
```

Example:



```
SELECT COUNT(*)
FROM employees;
```

This query returns the total number of rows in the employees table.

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5.4 SELECT IN

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The IN operator is used to filter the results based on a list of values. It checks if a value matches any value in a list or subquery.

Syntax:



```
SELECT column1, column2
FROM table_name
WHERE column_name IN (value1, value2, ... );
```

Example:

```
SELECT first_name, last_name
FROM employees
WHERE department IN ('Sales', 'Marketing');
```

This query retrieves employees who work in either the Sales or Marketing department.

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5.5 SELECT DATE

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The DATE function or type is used to work with date values. Its usage varies depending on the SQL database system. For instance, DATE can be used to extract the date part from a datetime column.

Syntax (for extracting date part):

```
SELECT DATE(column_name)
FROM table_name;
```

Example:

```
SELECT DATE(hire_date)
FROM employees;
```

This query extracts just the date part from the hire_date column.

5.6 SELECT SUM

The SUM function calculates the total of a numeric column.

Syntax:

```
SELECT SUM(column_name)
FROM table_name
WHERE condition;
```

Example:

```
SELECT SUM(salary)
FROM employees;
```

This query returns the total salary of all employees.

5.7 SELECT NULL

The keyword NULL represents missing or unknown data. You can use NULL in a SELECT statement to check for or retrieve rows with NULL values.

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Syntax:

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```
SELECT column1, column2
FROM table_name
WHERE column_name IS NULL;
```

Example:

```
SELECT first_name, last_name
FROM employees
WHERE manager_id IS NULL;
```

This query retrieves employees who do not have a manager (i.e., their manager_id is NULL).

Day 6:SQL Clauses

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